

Deepak Mallampati

APPLIED AI ENGINEER — FORWARD DEPLOYED

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH (open to SF relocation) -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building and shipping production Generative AI and LLM systems end-to-end — from messy enterprise data to deployed services running at scale. Embeds with stakeholders across product, platform, and data engineering, integrates production data pipelines and APIs, and owns features from architecture through deployment and iteration. Delivered **92% precision** RAG over millions of regulatory documents with a **40%** hallucination reduction, agentic orchestration cutting token consumption **42%**, and microservices sustaining **100,000+ requests daily** at **99.9% uptime**. Comfortable ramping into unfamiliar enterprise systems and customer environments, and translating ambiguous operational needs into working AI-native solutions.

CORE COMPETENCIES

Forward-Deployed / Customer-Embedded Delivery . End-to-End Ownership . Enterprise Systems & API Integration . Agentic LLM Systems . Production RAG & Retrieval . Production Data Pipelines (Kafka, Spark) . FastAPI Microservices . Python / TypeScript / SQL . Regulated-Domain Rigor (HIPAA / SOC 2 / PCI) . Technical Discovery with Stakeholders

EXPERIENCE

Progress Solutions

Jul 2025 -- Present

AI Engineer -- USA

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously — reducing token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 -- Jan 2025

AI Engineer Intern -- USA | Financial Services

- Developed AI agents and backend services for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.
- Optimized prompts and evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates **27%**; engineered content safeguards reducing unsafe outputs **42%**.

Emerson

Jun 2022 -- Apr 2023

ML Trainee -- India | Industrial / Operational Data

- Built regression and classification models on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting, with end-to-end preprocessing of raw sensor and operational datasets.
- Fine-tuned BERT and RoBERTa for entity extraction and text classification on domain-specific datasets, reaching **89% F1** on custom NER tasks.

PROJECTS

career-ops shipped -- in daily use

Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision. Founder-like end-to-end ownership — discovery, build, and ship of a self-running system that compresses hours of repetitive work into an automated workflow.

Python, Node.js, Claude Code, Playwright, multi-provider LLM orchestration, scheduled automation

MangaLens shipped -- Chrome Web Store

Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action. Full ship discipline: provider-agnostic vision abstraction, caching, scoped permissions, store review.

TypeScript, Manifest V3, content scripts, multi-provider vision APIs

Privacy-Preserving Clinical ML Pipeline

Framework for analyzing sensitive patient data without exposing identities — k-anonymity, l-diversity, and differential privacy — cutting re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance. Quantifies the privacy–utility trade-off across configurations.

Python, scikit-learn, differential privacy, IEEE-format research paper

EDUCATION

M.S. in Computer Science -- Kent State University, Ohio, USA

2025 | GPA 3.70 / 4.0

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React

GenAI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG architecture | multi-agent systems | structured outputs | fine-tuning (LoRA/QLoRA/PEFT) | evaluation (RAGAS, BERTScore)

Retrieval & Vector Search: FAISS, Pinecone, Weaviate | semantic & hybrid search | cross-encoder re-ranking | semantic caching

Integration & Data: REST APIs, microservices | PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | production data pipelines & orchestration

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow | observability/logging

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII masking/redaction | audit logging

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required